

module2_lab2

Module 2 Lab 2: Textbook Problem 4.2

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Course: CS82A - Introduction to Data Science with Python

GitHub Repository: <https://github.com/ramishatasfia5156-commits/cs82a-portfolio>

AI Disclosure: Claude (Anthropic) was used to help explain concepts and draft initial code for this lab. All code was reviewed, retyped, and understood by me. I can explain how both functions work and how the while loop terminates.

Part 1: Using a for Loop

```
[1]: def multiply_for(num1, num2):  
    """Multiply num1 by num2 using a for loop (repeated addition)."""  
    result = 0  
    for i in range(num2):  
        result += num1  
    return result  
  
# Test calls  
print("--- for loop version ---")  
print(f"7 x 6 = {multiply_for(7, 6)}")  
print(f"3 x 8 = {multiply_for(3, 8)}")  
print(f"12 x 5 = {multiply_for(12, 5)}")  
print(f"1 x 9 = {multiply_for(1, 9)}")  
print(f"0 x 4 = {multiply_for(0, 4)}")
```

```
--- for loop version ---  
7 x 6 = 42  
3 x 8 = 24  
12 x 5 = 60  
1 x 9 = 9  
0 x 4 = 0
```

Part 2: Using a while Loop

```
[2]: def multiply_while(num1, num2):  
    """Multiply num1 by num2 using a while loop (repeated addition)."""  
    result = 0  
    counter = 0  
    while counter < num2:  
        result += num1  
        counter += 1  
    return result  
  
    # Test calls  
    print("--- while loop version ---")  
    print(f"7 x 6 = {multiply_while(7, 6)}")  
    print(f"3 x 8 = {multiply_while(3, 8)}")  
    print(f"12 x 5 = {multiply_while(12, 5)}")  
    print(f"1 x 9 = {multiply_while(1, 9)}")  
    print(f"0 x 4 = {multiply_while(0, 4)}")
```

```
--- while loop version ---  
7 x 6 = 42  
3 x 8 = 24  
12 x 5 = 60  
1 x 9 = 9  
0 x 4 = 0
```

How the while Loop Terminates

The while loop ends due to the counter variable: This begins as 0 and increases by +1 with each cycle (counter+=1). The conditions for the while loop are that counter must be less than num2. Therefore when counter = num2 the while loop condition becomes False. Because Counter will ALWAYS increment by EXACTLY 1 during each cycle, it WILL eventually equal num2; thereby ending the loop within a finite amount of cycles.

Part 3: Verification : Both Functions Agree with *

```
[3]: test_cases = [(7, 6), (3, 8), (12, 5), (1, 9), (0, 4)]  
  
    print(f"{'num1':>5} {'num2':>5} {'for result':>12} {'while result':>14} {'*_  
        ↪result':>10} {'match?':>8}")  
    print("-" * 60)  
  
    all_correct = True  
    for num1, num2 in test_cases:  
        for_result = multiply_for(num1, num2)  
        while_result = multiply_while(num1, num2)  
        star_result = num1 * num2  
        match = (for_result == while_result == star_result)
```

```

    if not match:
        all_correct = False
    print(f"{num1:>5} {num2:>5} {for_result:>12} {while_result:>14}␣
↪{star_result:>10} {'YES' if match else 'NO':>8}")

print("-" * 60)
print(f"All results correct: {all_correct}")

```

num1	num2	for result	while result	* result	match?
7	6	42	42	42	YES
3	8	24	24	24	YES
12	5	60	60	60	YES
1	9	9	9	9	YES
0	4	0	0	0	YES

All results correct: True